

AVIK KUMAR DAS

Associate Professor, CSE (IoT, CS & BT) | AI-ML | Researcher

📍 Kolkata, India

✉ DSP Lab, Dept. of ETC, Indian Institute of Engineering Science And Technology, Shibpur, India- 711103

📞 9836820652 @ avikdas005@gmail.com

🐦 @005avik_das 🌐 avikdasetc

🔗 https://orcid.org/0000-0001-8824-703X



EXPERIENCE (6 YEARS)

Associate Professor (2 Years)

University of Engineering and Management

📅 March 2024 – Present

📍 UEM Kolkata

Visiting Professor (6 Months)

Calcutta University (CU)

📅 Sept 2023 – Feb 2024

📍 Calcutta University

Researcher (2 Years, Honorary)

Digital System Design (DSD Research Lab)

📅 Dec 2021 – January 2024

📍 Kolkata, India

Assistant Professor (1 Year 6 Months)

West Bengal University of Technology (MAKAUT)

📅 March 2017 – August 2018

📍 Seacom Engineering College

Examiner (1 Year)

West Bengal University of Technology (MAKAUT)

📅 July 2017 – August 2018

📍 WBUT, Kolkata, India

IT Support Executive (2 Years)

Goldstar Technology

📅 August 2012 – July 2014

📍 Kolkata, India

PUBLICATIONS (60) & PATENT (14)

Patent: 14 (Granted: 2, Published: 2, Filled: 10),
Journal: 10 (Published (P): 3, Accepted (A): 2, Communi-
cated (C): 8),

IEEE Transaction: 2 (Under review),

Book chapter: 11 (P:3, A:3, C: 5),

Conference: 37 (P: 15, A: 10, C: 12)

👤 Publications

- Das, A. K., & Pramanik, A. (2023). Image Transmission in UWA Channel using CS based OTFS system, Microsystem Technologies, Springer.
- Das, A. K., Pramanik, A., & Baksh, S. C. (2023). Convolution Neural Network for Sparse Channel and Image Reconstruction in Underwater Acoustic Communication, Taylor & Francis.
- Das, A. K., Pramanik, A., Bakshi, S. C., & Venkateswaran, P. (2023). Image transmission in underwater acoustic communication channel using LDPC codes. In *Proceedings of The International Conference on Computers and Devices for Communication (CODEC)*, IEEE Explore.
- Paul, A., Das, A. K., & et al. (2023). Development of Automated Cardiac Arrhythmia Detection Methods Using Single Channel ECG Signal. arxiv.2308.02405.

RESEARCH INTEREST

📡 Wireless communication and Signal processing: OTFS, voice and image processing.

📶 Underwater Acoustic Communication:

Channel estimation, modeling, OTFS, MIMO, Acoustic modem design, JANUS, PAMGUARD.

⚙️ LLM, GPT, ML, NLP and Deep Neural Network:

CNN, ML, NLP, LLM for data analysis & imbalanced data handling

🌐 Internet of Things (IoT) IoT, IoUT applications.

📶 Underground mine monitoring & positioning: LoRa, WSN.

💡 Visible Light Communication VLC: For application in GNSS denied areas

📶 Cognitive Radio Networks: Secure energy efficient and fast communication, WPT.

📶 Underground coal mine design: FISH

EDUCATION

🎓 PhD in Electronics and Telecommunica-
tion Engineering

IIST, Shibpur, Howrah

📅 Pre-submission of thesis completed on 13.09.2023

🎓 M.Tech. in Information and Communica-
tion Engineering

IIST, Shibpur, Howrah

📅 June 2014 – May 2016, 60.67%

🎓 B.Tech. in Electronics and Communication
Engineering

West Bengal University of Technology

📅 August 2008 – June 2012, 8.13

🎓 Higher Secondary (10+2) Math Phy Chem
Bio

WBCHSE, Rajbalhat High School

📅 May 2007, 78.80%

🎓 Secondary (10 Madhyamik)

WBBSE, Rajbalhat High School

📅 May 2005, 79.12%

HONORS AND AWARDS

🏆 Received PRISM Funding, DSIR, Govt. of India,
by IIT Kgp, 2024

🏆 Received MeitY TIDE Fund, Govt. of India,
by SINE IIT B, 2023

- Avik Kumar Das, Ankita Pramanik, Ankita Ray Chowdhury, Saikat Chandra Bakshi, Venkateswaran, Palanian-davar. (2024). A System for Locating Mine Personnel in an Underground Coal Mine and Method Thereof Remote Positioning System for Underground Coal Mines using Visible Light and LoRa Communication Visible Light-Based Location Tracking System for Enhancing Miner Rescue and Monitoring. Indian Patent, Application No. 202441028230.
- Avik Kumar Das, Ankita Pramanik, Ankita Ray Chowdhury, Saikat Chandra Bakshi, Venkateswaran, Palanian-davar and Suvankar Bose. (2024). An Internet of Underwater Things (IoUT)- based water quality monitoring system and method of operation. Indian Patent, Application No. 202441044738 A.
- Avik Kumar Das, et al. (2024a). Enhanced Convolutional Neural Network for Sign Character Recognition with Mixed Pooling and Mish Activation Function. In *Track 4: Biomedical instrumentation, ieee conference*. Published.
- Avik Kumar Das, Shaoni Banerjee, Hansa Ray, Shubhan-shu Tiwari, Khizra Aslam, Apurba Nandi. (2024). Enhanced Stock Price Forecasting Using a Hybrid Bidirectional LSTM-GRU Model: Addressing Volatility and Time Dependencies in Time Series Data. In *Proceedings of irtm 2024*. Accepted and Presented.
- Das, A. K., & et. al. (2024). Enhanced Convolutional Neural Network for Sign Character Recognition with Mixed Pooling and Mish Activation Function. In *2024 International Conference on Intelligent Systems and Advanced Applications (ICISAA)*. IEEE, Pune.
- Avik Kumar Das, Ankita Pramanik, Ankita Ray Chowdhury, Saikat Chandra Bakshi, Venkateswaran, Palanian-davar. (2023a). Smart auto charging AUV for water quality estimation with dehazed photo capturing. Indian Patent, Application No. 202331063691.
- Avik Kumar Das, Ankita Pramanik, Ankita Ray Chowdhury, Saikat Chandra Bakshi, Venkateswaran, Palanian-davar. (2023b). Smart underground mine monitoring system. Indian Patent, Application No. 202331063085.
- Das, A. K., & Pramanik, A. (2022). Efficient Image Transmission in UWA Channel. In *2022 IEEE Ocean Engineering Technology and Innovation Conference: Management and Conservation for Sustainable and Resilient Marine and Coastal Resources (OETIC)* (pp. 55–61). IEEE.
- Das, A. K., Pramanik, A., Chowdhury, A. R., & Ramakrishnan, L. (2022). On Improved Performance of Underwater VLC System. In *Proceedings of The International Conference on Innovation in Technology (INOCON)*, IEEE, Bangalore.
- Das, A. K., & Pramanik, A. (2021). A Survey Report on Underwater Acoustic Channel Estimation of MIMO-OFDM System. In *Proceedings of International Conference on Frontiers in Computing and Systems: COMSYS 2020* (pp. 745–753). Springer.
- Das, A. K., & Pramanik, A. (2020). A Survey Report on Deep Learning Techniques in Underwater Acoustic Channels. In *Computational Intelligence in Pattern Recognition* (pp. 407–416). Springer.
- Pramanik, A., & Das, A. K. (2020). A Novel High rate LDPC code. In *Proceedings of the 21st International Conference on Distributed Computing and Networking (ICDCN)* (pp. 1–4).

- Fund for IIC Prototype competition project at IEST, Shibpur, 2022 to present
- Winner of IEEE R10 WIE Hackathon, On Smart Resource Management, 2023, Jadavpur University, Kolkata
2022 to present
- Prize in photography competition by Botanical Survey of India (BSI) at AJC Bose Indian Botanic Garden, 2022, 2023
- Photographer of AJC Bose Indian Botanic Garden (BSI) 2022 to present
- Speaker in the webinar on Realization and application of low pass filters using FPGA
Organized by Digital System Design · Jan 2022
- BFA (Bachelor of Fine Arts) Surabharati Sangeet Parishad, Kolkata
- Goalkeeper of IEST, Shibpur. 2014 to 2016, 2018 to present
- Speaker in the IIC seminar on IoT and Embedded system at BIT Techno India, Kolkata. 2023
- Life Member of OSI, MSI
- Nominated and Shortlisted in the Top 3 in the Globe for the Final Round of AEGIS GRAHAM BELL Award
Project Title: Smart Water Monitoring and Management System for Aquatic Environments Using IoT and Machine Learning.

PROJECTS

- 1. Underwater VLC and UWA-based UUV design for underwater environment exploration: IEST, Shibpur, August 2019 to present.
- 2. IoT and LoRa-based Underground Mine Monitoring and Positioning System: IEST, Shibpur, October 2020 to present. (IBM)
- 3. IoT and UV-based Water Quality Monitoring floating node and WPT based docking station: IEST, Shibpur, Feb 2021 - Present
- 4. Real-Time Visibility Enhancement: IEST, Shibpur, July 2021 to present (PRISM funding).
- 5. NLP and LLM-based data analysis & voice, speech processing and alerting system: Addauto Technology Pvt. Ltd., July 2021 to present.
- 6. Smart agriculture system for field, greenhouse, precision farming, & hydroponics applications: Addauto Technology Pvt. Ltd., July 2021 to present.
- 7. Smart & green cafe: BECC, at IEST, Shibpur, July 2023 to present.
- 8. 6G channel modeling and terahertz communication using AI. January 2024 to present.
- 9. Energy Efficient and Secured Spectrum Sensing in Cognitive Radio Networks Energy Efficient and Secured Spectrum Sensing in Cognitive Radio Networks: IEST, Shibpur, (M.Tech project) Oct 2014 - May 2016.
- 10. Miniature FM transmitter: BUIE (B.Tech project) Oct 2011 - May 2012.
- 11. PI: GAPPS: GNSS and AI-Based Precision Positioning System. UEM Kolkata. Sept 2024 - Aug 2025.
- 12. PI: Track your Green Ride (TYGR): An IoT and AI Enabled Advanced Real-Time Vehicle Tracking system Using GNSS and Health Monitoring for Safety and Sustainability. UEM Kolkata. Sept 2024 - Aug 2025.
- 13. PI: SMART-ARAI: Smart Traffic Optimization and Routing System with Artificial Intelligence and

TRAINING AND COURSES

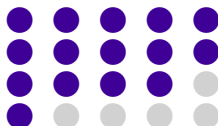
- ◆ Concepts and applications of GIS, remote sensing and GNSS: SERB High-End Workshop (KARYASHALA), NIT Jamshedpur, July 2023
- ◆ 17th National Frontiers of Engineering (NatFoE23): BIT, Meshra, June 2023
- ◆ Enabling Technologies for Electric Transportation: by MNIT Jaipur and Concordia University, Canada
- ◆ Cyberphysical Digital Microfluidic Biochips in Healthcare Technologies and Applications: IIST, Shibpur, SPARC, Sponsored
- ◆ Digital Humanities: IIST, Shibpur
- ◆ GIAN (Global Initiative for Academic Networks) Course: IIST, Shibpur
- ◆ Training in BSNL on Wireless and Mobile Communication, Networking.: NSCBTTC, Kalyani, West Bengal
- ◆ FDP attended 10 on AI, IoT and Cloud computing.
- ◆ Courseara courses 10 on AI, IoT and Cloud computing

SKILLS

- Coding: Python, R, SQL, C, HTML, RTKLib
- Publishing: LATEX
- FPGA, Matlab, LabVIEW, TMS3200C6748
- CCS, Photoshop, UX UI design
- Arduino, ESP, Eagle PCB, R Pi, LattePanda

LANGUAGES

English
Bengali
Hindi
Sanskrit



REFERENCES

◆ Prof. Ankita Pramanik
◆ Indian Institute of Engineering Science And Technology, Shibpur
@ ankita@telecom.iests.ac.in

◆ Prof. Santi Prashad Maity
◆ Indian Institute of Engineering Science And Technology, Shibpur
@ santipmaity@it.iests.ac.in

◆ Prof. Chinmoy Ghorai
◆ Jadavpur University, Kolkata
@ cghorai.etce@jadavpuruniversity.in

◆ Prof. Gopal Chandra Roy
◆ Indian Institute of Engineering Science And Technology, Shibpur
@ gcr@mining.iests.ac.in

V2V Communication . UEM Kolkata. Sept 2024 - Aug 2025.

◆ 14. PI: AQUA-SENSE- AI: Advanced Quantitative Underwater Monitoring & Sensor-Enhanced Networked Systems for the Internet of underwater Things (IoUT) applications. UEM Kolkata. Sept 2024 - Aug 2025.

◆ 15. PI: AI-PLANT-VISION: AI-based Predictive Learning and Advanced Neural Tools for Vision-Based Disease Monitoring. UEM Kolkata. Sept 2024 - Aug 2025.

◆ 16. PI:IoT and AI -Enabled Healthcare Monitoring System Using Active Antennas for Real-Time Data Collection and Analysis. Sept 2024 - Aug 2025.

RESPONSIBILITIES

◆ Member of R&D team of of Digital System Design Lab. and Addauto Technology Pvt. Ltd. Kolkata 2021 to present.

◆ Member of Cine Production of House & Studio, Kolkata. 2021 to present.

◆ Volunteer of Higher Education Conclave, Digital Humanity conferences. 2019, IIST, Shibpur.

◆ Coordinator of Student research and Grant in projects 2025, UEM Kolkata, India.

TEACHING ASSISTANTSHIP

◆ Digital Image Processing and Computer Vision Lab: Matlab, Python, CCS, TMS320C6748, IIST, Shibpur, 2018 to present.

◆ Digital Signal Processing Lab: Matlab, CCS, TMS320C6748, IIST, Shibpur, 2018 to present.

◆ Communication Engineering Lab: IIST, Shibpur, 2015-2016. Digital communication

◆ Computer Network Lab: IIST, Shibpur, 2015 - 2016.

◆ Operating system Lab: IIST, Shibpur, 2015 - 2016.

SUBJECTS & LABS.

◆ Wireless communication, Computer networks, Analog circuit, DSP, EM Theory, Seacom Engineering College, 2017-2018. Digital communication

◆ AI ML, IoT, DS and Algo, Computer Architecture, Cloud Computing and IoT, Signal Processing UEM K, 2024

PERSONAL DETAILS

◆ Father's Name: Mr. Dipak Kumar Das

◆ Mother's Name: Mrs. Anita Das

◆ Permanent Address: Village:Mukundapur, P.O :- Rajbalhat

District: - Hooghly, PIN: 712408, West Bengal, India (Tarakeswar)

◆ Date of Birth: 21-12-1989

◆ Gender: Male.

◆ Marital Status: Single.

◆ Hobbies and interests: Football, Playing guitar, Photography, Painting, Short film making, Prototype design, Bike riding

🎓 **Prof. Monojit Mitra**

📍 Indian Institute of Engineering Science And Technology,
Shibpur

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🎓 **Sourav Bhattacharyya**

📍 Senior Data Scientist, IBM, India

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🎓 **Prof. Hafizur Rahaman**

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Anik Kumar Das.

I hereby declare that the above written particulars are true to the best of my knowledge and belief.